

Lecture 精哥某朝 supper court 的卡位。

由于规则为猜测得为个位数，而该得为本身概率极不公平！

将 Page 761 #769 中的 10 更改，基于一个平均为基础中所得，但公众本身也具猜以意。虽然好的在于得钱概率大，但人也多，年分下来钱很少，所以需将利率提高，即“最不利下注意”

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只需找到该括号的真实发生概率远大于公众认为其发生的概率即可。

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Chapter 18 Random Variables

was selected by a large fraction of the population. Apparently many people think the same way; they pick the same numbers not on purpose as in the previous game with Nick and Eric, but based on the Red Sox winning average or today's date. The result is as though the players were intentionally colluding to lose. If any one of them guessed correctly, then they'd have to split the pot with many other players. By selecting numbers uniformly at random, Chernoff was unlikely to get one of these favored sequences. So if he won, he'd likely get the whole pot! By analyzing actual state lottery data, he determined that he could win an average of 7 cents on the dollar. In other words, his expected return was not $-\$0.50$ as you might think, but $+\$0.07$.² Inadvertent collusion often arises in betting pools and is a phenomenon that you can take advantage of.

★ 线性性不依赖于结构 ★

18.5 Linearity of Expectation

Expected values obey a simple, very helpful rule called *Linearity of Expectation*. Its simplest form says that the expected value of a sum of random variables is the sum of the expected values of the variables.

Theorem 18.5.1. For any random variables R_1 and R_2 ,

$$\text{Ex}[R_1 + R_2] = \text{Ex}[R_1] + \text{Ex}[R_2]$$

Proof. Let $T ::= R_1 + R_2$. The proof follows straightforwardly by rearranging terms in equation (18.2) in the definition of expectation:

$$\begin{aligned} \text{Ex}[T] &::= \sum_{\omega \in S} T(\omega) \cdot \text{Pr}[\omega] \\ &= \sum_{\omega \in S} (R_1(\omega) + R_2(\omega)) \cdot \text{Pr}[\omega] && (\text{def of } T) \\ &= \sum_{\omega \in S} R_1(\omega) \text{Pr}[\omega] + \sum_{\omega \in S} R_2(\omega) \text{Pr}[\omega] && (\text{rearranging terms}) \\ &= \text{Ex}[R_1] + \text{Ex}[R_2]. && (\text{by (18.2)}) \end{aligned}$$

A small extension of this proof, which we leave to the reader, implies

²Most lotteries now offer randomized tickets to help smooth out the distribution of selected sequences.

72] lecture and other implications

Linearity of Expectation

于无项时的求和条件：

期望线性方程并不适用于无穷项的情况：

72] 只有 $\sum_{i=0}^{\infty} \text{Ex}[R_i] < \infty$ 时才

成立，即：所有随机变量绝对值的期望之和必须有限。

$$\text{E[payoff]} = \sum (\text{Outcome} \times \text{Pr}(\text{Outcome}))$$